# Genetics in Colorectal Cancers

## Genetics of Colorectal Cancer

| Gene | Associated Syndrome or Cancer(s) | Overall Penetrance | Patients With Mutation, No. (%) | (95% CI) |
|---|---|---|---|---|
| Any pathogenic or likely pathogenic mutation | | | 72 (16) | (12.8-19.8) |
| Genes associated with colon cancer | | | 59 (13.1) | (10.2-16.7) |
| *MLH1* | Lynch syndrome | High | 13 (2.9) | (1.6-5.0) |
| *MSH2* | Lynch syndrome | High | 16 (3.6) | (2.1-5.8) |
| *MSH2*/monoallelic *MUTYH* | Lynch syndrome/colon cancer | High/low | 1 (0.2) | (0.01-1.4) |
| *MSH6* | Lynch syndrome | Moderate | 2 (0.4) | (0.08-1.8) |
| *PMS2* | Lynch syndrome | Moderate | 5 (1.1) | (0.4-2.7) |
| *APC* | Familial adenomatous polyposis (FAP) | High | 5 (1.1) | (0.4-2.7) |
| *APC p.I1307K* | Colon cancer | Low | 4 (0.9) | (0.3-2.4) |
| *MUTYH* | | | | |
| Biallelic | MUTYH-associated polyposis (MAP) | High | 4 (0.9) | (0.3-2.4) |
| Monoallelic | Colon cancer | Low | 7 (1.6) | (0.7-3.3) |

JAMA Oncol March 31, 2017

## FAP screening

Colonoscopy age 10-12 EGD for duodenal polyps at age 20-30 CT 1-3 years after colectomy and q5 yers in those with family hx of desmoids

## Lynch Syndrome

- 3% of all colorectal cancers
- Germline mutations of mismatch repair genes
  - *OR* Germline deletions in EPCAM gene
- 80% lifetime risk colorectal cancer
- Early onset ($\tilde{x}$ 45-60yrs)

- Tumors tend to be right-sided

## Lynch Non-Colorectal Cancers

A Comprehensive Framework for Early-Onset Colorectal Cancer Research. The Lancet. Oncology. 2022. Eng C, Jácome AA, Agarwal R, et al.

- Endometrial (25-60% penetrance)
- Stomach
- Ovarian (45-125??? penestrance)
- Urinary tract
- Biliary Tract
- Small bowel
- Male breast cancer (1.2% penetrance vs 0.1%)

(**sinicrope764?**)

## Lynch Screening

Colonoscopy q1-2 years staring age 20-25

## Mismatch Repair Protein Loss in Lynch Syndrome

Lynch Syndrome can be caused by loss of expression of:

- MLH1
- PMS1
- MSH6
- MSH2
- MSH5

## Mismatch-repair Deficient Colorectal Cancers

Deficiency in mismatch repair proteins → microsatellite instability

15% of colorectal cancers are MMR-deficient

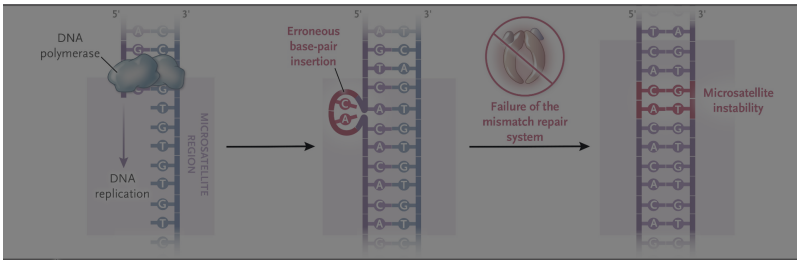
- Sporadic - Promoter methylation of MLH1 by BRAF V600E mutation (10%)

- Lynch - Due to mutations in mismatch repair genes (5%) 90% of Lynch tumors are MMR-deficient

*Reflex testing for BRAF V600E mutation is essential for MMR-deficient tumors*

## Mismatch Repair Proteins and Microsatellite Instability

Deficiency in mismatch repair proteins → microsatellite instability




## Mismatch Repair and MSI Testing

MMR Protein expression can be detected by immunohistochemistry

- Routine testing of tumor for GI cancers at CMC
- Loss of MLH1 leads to loss of PMS2

Microsatellite Instability can be detected by pCR

- Tumors are not routinely tested for MSI at CMC

97-99%+ concordance between dMMR and MSI-High status

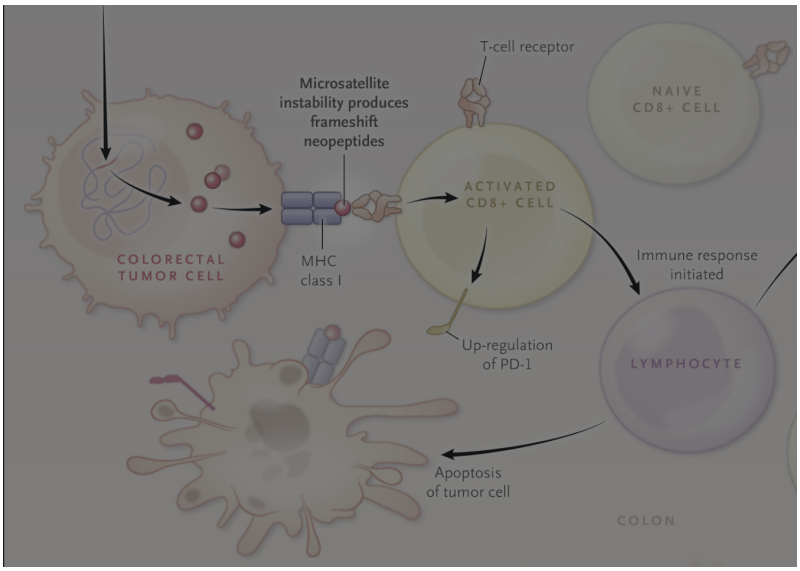
## MMR-deficient Tumor Clinical Feaures

- Proximal location
- Poor differentiation
- Mucinous histology
- Better prognosis (in early stage)
- Respond well to immunotherapy (ICI)

- Less responsive to 5-FU chemotherapy (in stage II)

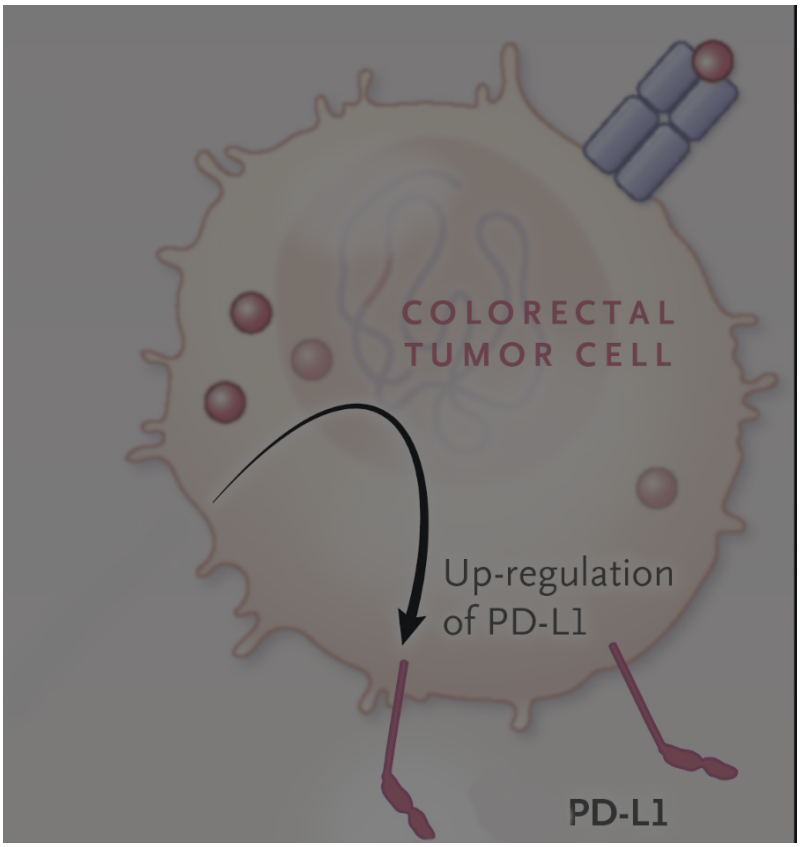
## MMR-deficient Tumors are Immunogenic

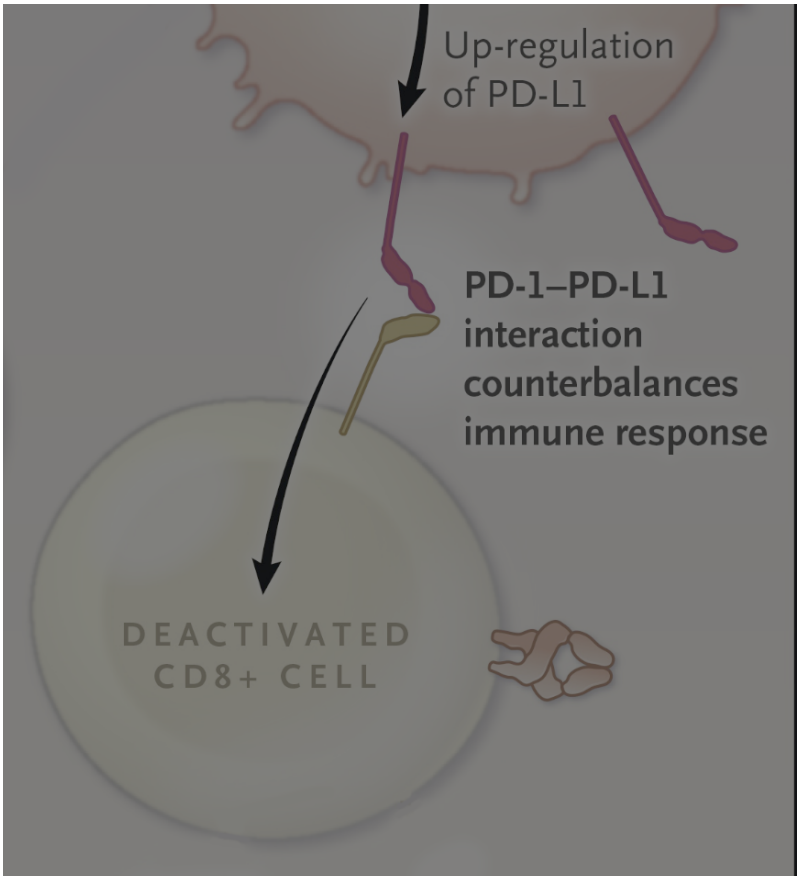
Mutational burden from dMMR causes MSI-High tumors to be immunogenic



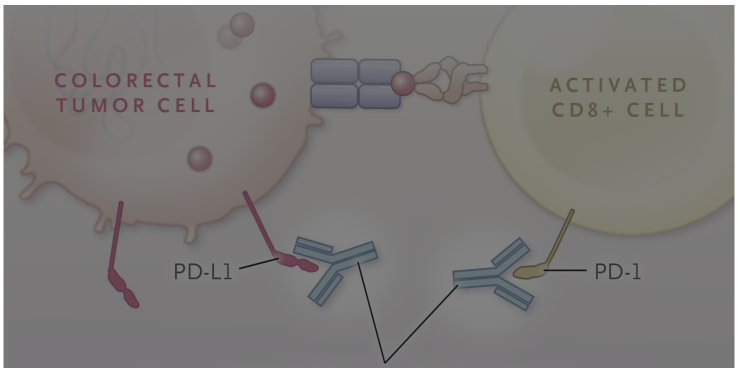

Deficient Mismatch Repair/Microsatellite Unstable Colorectal Cancer: Diagnosis, Prognosis and Treatment. European Journal of Cancer. 2022. Taieb J, Svrcek M, Cohen R, et al.

**Tumor PD-L1 Down-regulates Immune Responses**






**Immune Checkpoint Inhibitors**




Inhibition of PD-L1 and PD-1 interaction activates T cells

## PD-L1 expression is higher in MMR-deficient Colorectal Cancers

Immunohistochemistry analysis of 1800 colorectal cancer specimens:

PD-L1 was more often positive in dMMR (18.6%) than in MMR proficient (pMMR) cancers (4.1%; $p < 0.0001$)

(**moller1210?**)

## MSI-High Colorectal Cancers

Clinical Relevance of Microsatellite Instability in Colorectal Cancer. Journal of Clinical Oncology : Official Journal of the American Society of Clinical Oncology. 2010. de la Chapelle A, Hampel H.

## BRAF V600E Colorectal Cancer Features

1. Age >70
2. Female
3. Proximal Right sided tumors
4. MLH1 hypermethylation
5. High grade and poor differentiation
6. More peritoneal and nodal metastasis
7. Less lung metastais

## Neoadjuvant Rx for MMR-deficient Colon Cancer - NICHE

115 colon cancer: neoadjuvant nivolumab + ipilimumab /*rightarrow* surgery

109 patients had a pathological response

105 Major pathological respoinse (TRG1 or TRG2)

- 75 pathological complete response

No recurrence of disease at median follow up of 26 months

(Chalabi et al. 2024)

## Adjuvant Therapy Stage II MMR-deficient Colon Cancer

ASCO: No benefit of treatment of stage II dMMR colon cancer with 5-FU alone

Ppooled analyses showing no difference in disease-free survival (HR 0.89, 95% CI 0.45-1.74) or overall survival (HR 1.03, 95% CI 0.53-2.02) with adjuvant chemotherapy in this population.

(**ribic247?**)

## Adjuvant Therapy of Stage III MMR-deficient Colon Cancer

Oxaliplatin-based chemotherapy (FOLFOX or CapeOX) and atezolizumab (Tecentriq)

ATOMIC trial: atezolizumab + mFOLFOX6 is superior to mFOLFOX6 alone, with a 3-year disease-free survival rate of 86.4% versus 76.6% (hazard ratio 0.50, P .0001)

## Colon Polyposis: >10 adenomatous polyps

- Classical FAP
- Attenuated FAP (AFAP)
- MUTYH-associated polyposis (MAP)
- Colonic adenomatous polyposis of unknown etiology (CPUE)

## Colon Polyposis: >4 hamartomatous polyps

- Puetz-Jaghers
- Juvenile Polyposis Syndrome
- Cowden Syndrome/PTEN Hamartoma Tumor Syndrome

## Serrated Colon Polyps

5 serrated polyps/lesions proximal to the rectum, all being 5 mm in size, with 2 being 10 mm in size OR >20 serrated polyps/lesions of any size distributed throughout the large bowel, with 5 being proximal to the rectum

## FAP

(FAP) is caused by germline mutations in the APC gene and leads to the development of hundreds to thousands of colorectal adenomas, with nearly 100% risk of colorectal cancer if untreated. [4]

## Attenuated FAP

~30 polyps 70% penetrance by age 80 - mean age at dx 50-55

## References

Chalabi, Myriam, Yara L. Verschoor, Pedro Batista Tan, et al. 2024. "Neoadjuvant Immunotherapy in Locally Advanced Mismatch Repair-Deficient Colon Cancer." *The New England Journal of Medicine* 390 (21): 1949–58. https://doi.org/10.1056/NEJMoa2400634.